Pulicherla Pavan Kumar Reddy

pavankumarreddy3364@gmail.com | **+91 6301596682** | **Bengaluru** | **LinkedIn**

SUMMARY

Full-Stack Software Engineer with 2.9+ years of experience building scalable web applications using Django, Django REST Framework, and Angular. Skilled in designing RESTful APIs, optimizing MySQL databases, and implementing automated workflows and web scraping solutions using Playwright. Strong focus on clean architecture, modular code, performance optimization, and delivering end-to-end features across backend and frontend systems.

Skills

Languages: Python, SQL

Databases: MySQL

Web & Automation Tools: Playwright, Selenium

Libraries: Matplotlib, Numpy and Pandas

Frameworks & Libraries: Django, Django REST Framework, Angular

API Testing & Documentation: Postman, Swagger

DevOps & Deployment: AWS, Git, GitHub, Docker, Gunicorn, Nginx

Experience

Python Backend Developer *(Full time)* | ***IIS – Hyderabad*** **Mar 2025 – Present**

- Refactored multiple Django generic views (List, Retrieve, Update) into a unified **NautobotUIViewSet**, reducing boilerplate and improving maintainability across plugins.
- Developed reusable base viewsets and mixins to centralize CRUD logic, ensuring consistency and reducing code duplication across multiple Nautobot apps.
- Integrated Nautobot's UI Component Framework for declarative UI panel creation, enabling dynamic and consistent rendering of plugin views and templates.
- Documented UI behavior before and after refactoring, providing clear comparative analysis that demonstrated improvements in consistency, performance, and user experience resulting from the new NautobotUIViewSet architecture.

Python Full Stack Developer *(Full time)* | ***Datasteer – Bengaluru*** **Dec 2022 – Dec 2024**

- Developed comprehensive ERP modules, implementing backend logic using Django models and ensuring seamless integration with Angular-based frontend interfaces.
- Built and optimized RESTful APIs for managing complex operations such as sale order creation, updates, and retrieval, enabling smooth data flow between components.
- Implemented custom save methods for sale orders to manage dynamic workflow transitions, including parent-child order status handling and automated flow stage updates.
- Utilized APIViews for detailed logic handling and ViewSets for efficient lookup table management, enhancing modularity and scalability.
- Implemented advanced filtering mechanisms in Django REST Framework (DRF) by creating a dedicated filters.py file, integrating custom filters for efficient data querying and dynamic API responses, enabling granular data retrieval based on user-defined criteria.
- Developed a user registration system with OTP-based verification (email/SMS), implementing secure OTP generation, validation, and expiration logic to ensure authenticated user onboarding.
- Designed and developed complex RESTful APIs using APIViews, combining data from multiple relational tables into a consolidated JSON payload to streamline communication between the backend and Angular frontend, reducing API calls and improving performance.
- Designed and implemented a secure authentication system using JWT (JSON Web Tokens), including login, token generation, and route protection. Integrated token-based authorization to ensure secure API communication and enhanced overall application security.

Projects

Nautobot | *Django, Python* **Mar 2025 – Present**

- **Description:** Nautobot is an open-source Network Source of Truth and Automation Platform developed by Network to Code (NTC). It extends NetBox and provides a flexible framework for network data modeling, orchestration, and plugin-based UI/API customization.
- **Contribution:** Developed custom Nautobot apps and UI ViewSets using Django and the Nautobot UI Component Framework, refactoring legacy generic views into modular, reusable components for enhanced performance and maintainability.

ERP Software Development | *Django Rest Framework, Angular, MySQL* **Mar 2022 – Dec 2024**

- **Description:** Developed a full-scale ERP system integrating Django backend services with an Angular frontend, enabling streamlined business workflows, automated order processing, and efficient data management across modules such as sales, invoicing, product workflows, and customer history.
- **Contribution:** Led backend development by building and optimizing RESTful APIs, designing complex multi-table data aggregation using APIViews, implementing custom save methods for dynamic workflow transitions, and creating advanced DRF filtering through dedicated custom filter classes. Engineered automated backend processes for order handling, invoicing, and product creation, ensuring scalable, maintainable, and high-performance data operations throughout the ERP.

Performance Enhancements

- Optimized API performance by implementing efficient pagination strategies in Django REST Framework using LimitOffsetPagination and PageNumberPagination, reducing API response times and enhancing performance for data-heavy endpoints.
- Designed and implemented complex RESTful APIs in Django REST Framework, consolidating data from multiple relational tables into a single optimized payload, improving frontend data loading speed by 25% and enhancing overall API efficiency.
- Enhanced frontend performance by developing reusable Angular components and optimizing data binding techniques, reducing page load times by 20% and improving user experience across ERP modules.
- Improved MySQL database performance by optimizing complex JOIN queries, implementing proper indexing strategies, and fine-tuning query execution plans, achieving a 35% reduction in query execution time.
- Implemented advanced search and filtering in Django REST Framework using `django-filter` and custom filter backends, enabling dynamic querying and significantly improving data retrieval efficiency.

Education

SASTRA DEEMED UNIVERSITY **June 2018 – June 2022**
Bachelor of Technology in Computer Science and Engineering (CGPA of 7.06)
(Thanjavur, India)

Sri Chaitanya Junior College **May 2016 – June 2018**
MPC (Percentile of 94.5)
(Tirupati, India)